

Clinical Risk Assessment Report: Patient 50

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 42.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.93x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.2%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.66 [Top 14%] (+5.2%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.6 [Bottom 18%] (+3.3%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.3 [Bottom 24%] (+3.2%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.69 [Top 23%] (+2.7%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.9 [Top 19%] (+2.7%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.67 [Avg (67th %ile)] (+2.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 15%] (+2.1%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.06 [Top 15%] (+1.9%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

